



Student Number:.....

**Gosford High School
2019 Higher School Certificate
Trial Examination**

Chemistry

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- NESA-approved calculators may be used
- Three data sheets and a Periodic Table are provided at the back of this paper

Total marks: 100

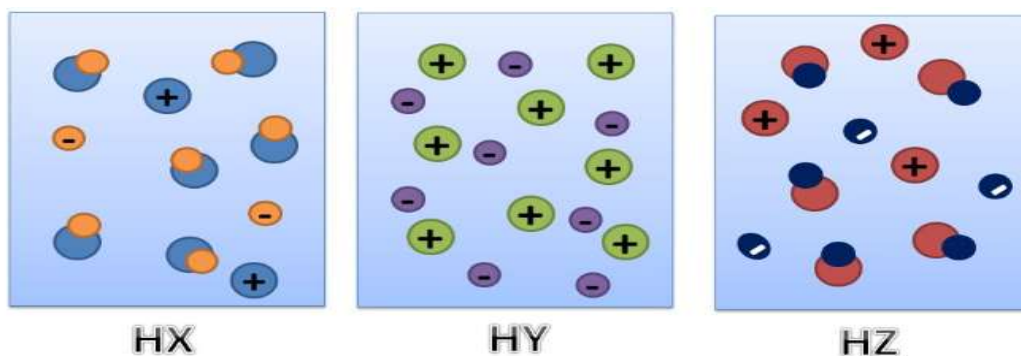
Section I – 20 marks

- Attempt Questions 1-20
- Allow about 35 minutes for this section

Section II – 80 marks

- Attempt Questions 21-35
- Allow about 2 hours and 25 minutes for this section

- 3 Three solutions of acids, HX, HY and HZ, are represented by the following diagrams. For clarity, water molecules have not been shown.



The acids, in order of increasing acid strength, are

- A. HX, HY, HZ
 - B. HZ, HX, HY
 - C. HX, HZ, HY
 - D. HY, HZ, HX
- 4 Consider the following organic compounds:

I – $\text{CH}_3\text{CH}_2\text{COOH}$

II – CH_3COCH_3

III – $\text{CH}_3\text{CH}_2\text{CHO}$

Which of the following statements is correct?

- A. Compounds I, II and III belong to the same homologous series.
- B. Compounds II and III have the same molecular formula.
- C. Compound I has the lowest boiling point.
- D. Compounds I and III would combine to form an ester and water in the presence of concentrated sulfuric acid.

5 Which of the following shows the product(s) of reaction of but-1-ene with water, in the presence of an acid catalyst?

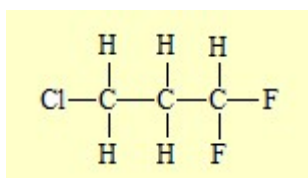
- A. Butan-1-ol only.
- B. Butan-2-ol only.
- C. A mixture of butan-1-ol and butan-2-ol.
- D. A mixture of butan-1-ol and butan-4-ol.

6 The pH of a $0.0001 \text{ mol L}^{-1}$ solution of a monoprotic acid was measured by a student and found to be 4.

What proportion of the acid molecules has been converted to ions?

- A. 0%
- B. 4%
- C. 96%
- D. 100%

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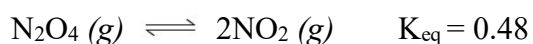
The compound shown above is

- A. 2-chloro-3,3-difluoropropane
- B. 3-chloro-1-fluoropropane
- C. 1-chloro-3,3-difluoropropane
- D. 3-chloro-1,1-difluoropropane

8 In which of the following pairs are the substances isomers of each other?

- A. 1-pentene and cyclopentane
- B. Pentane and 2,3-dimethylbutane
- C. Butanoic acid and 1,2-butanediol
- D. Dichloromethane and trichloromethane

9 At 100°C, the equilibrium constant for the reaction below is 0.48.



In an experiment it was found that the equilibrium concentration of $\text{N}_2\text{O}_4 (g)$ was 0.20 mol/L. Calculate the concentration of $\text{NO}_2 (g)$ in this equilibrium mixture, in mol/L.

- A. 0.096
- B. 0.31
- C. 0.24
- D. 0.48

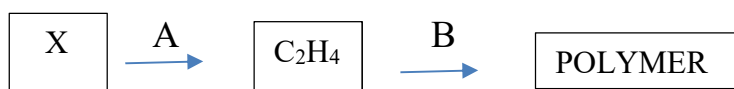
10 Which is the INCORRECT statement about the reaction when liquid ethanol burns in air?

- A. The enthalpy and entropy changes for the reaction are both negative.
- B. The reaction is described as combustion.
- C. The reaction does not reach equilibrium and the system is classified as open.
- D. The enthalpy change is negative and gases form.

11 The low boiling point of liquid hydrocarbons are often classified as being volatile. This means that:

- A. they can ignite, even without an ignition source.
- B. they can easily change into gas at room temperature.
- C. their products of combustion are toxic.
- D. they all give out the same amount of energy when combusted.

12 Which of the following most correctly represents X, A,B in the flow chart?



- | | X | A | B |
|----|---------|-------------|-----------------------------|
| A. | Glucose | dehydration | addition polymerisation |
| B. | Ethanol | oxidation | condensation polymerisation |
| C. | Glucose | addition | condensation polymerisation |
| D. | Ethanol | dehydration | addition polymerisation |

13 The table gives heat of combustion kJ g⁻¹ for a number of different fuels.

Fuel	Heat of combustion kJ g ⁻¹
Butanol	30.8
Pentanol	36.5
Hexanol	41.2
Petrol (Octane)	47.8

The heat of combustion in kJ mol⁻¹ for one of the fuels was calculated as 3218 kJ mol⁻¹. What was the fuel?

- A. Hexanol
- B. Octane
- C. Butanol
- D. Pentanol

- 14 The table below shows pK_{ind} , the pH range, and the colour changes of three indicators.

Indicator	pK_{ind}	pH range	colour at lower pH	colour at higher pH
bromophenol blue	4.0	3.0 - 4.6	yellow	blue
methyl red	5.1	4.2 - 6.3	red	yellow
phenolphthalein	9.3	8.3 - 10.0	colourless	red

At a pH of 4 which option below correctly identifies the colour of the solution:

	Bromophenol blue	Methyl red	Phenolphthalein
(a)	yellow	red	colourless
(b)	Yellow	yellow	colourless
(c)	Green	orange	colourless
(d)	Green	red	colourless

- 15 What is the pOH of the solution formed when 20.00 mL of 0.250 mol/L HCl is mixed with 80.00 mL of 0.350 mol/L NaOH solution at 25°C?
- A. 0.638
- B. 1.638
- C. 12.362
- D. 13.362

- 16 In a titration, a 25.00 mL titre of 1.00 mol L⁻¹ hydrochloric acid neutralised a 20.00 mL aliquot of sodium hydroxide solution which was in a conical flask. If, in repeating the titration, a student failed to rinse one of the pieces of glassware with the appropriate solution, the titre would be:
- A. greater than 25.00 mL, if the final rinsing of the 20.00 mL pipette was with water rather than the base.
 - B. less than 25.00 mL, if the final rinsing of the burette was with water rather than the acid.
 - C. equal to 25.00 mL, if water was left in the conical flask after final rinsing.
 - D. greater than 25.00 mL, if the conical flask was rinsed with the acid prior to the addition of the aliquot.
- 17 Which of the following will cause the greatest change in pH to a 100 mL sample of 0.1 mol L⁻¹ HCl?
- A. adding 1.0g of calcium carbonate powder
 - B. adding 10.0 mL of 0.1 mol L⁻¹ NaOH
 - C. adding 100.0 mL of 0.1 mol L⁻¹ HCl
 - D. diluting the sample to 1000.0 mL
- 18 The equilibrium constant for the reaction below has an equilibrium constant K_I .
- $$\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g}) \quad K_I = 159 \text{ at } 500\text{K}$$
- At the same conditions of temperature and pressure, what is the equilibrium constant for the reaction:
- $$\text{HI}(\text{g}) \rightleftharpoons \frac{1}{2}\text{H}_2(\text{g}) + \frac{1}{2}\text{I}_2(\text{g})$$
- A. 0.00629
 - B. 0.0793
 - C. 12.6
 - D. 79.5

- 19** When a sample of $\text{NO}_2(g)$ is placed in a container, the following equilibrium is rapidly established:



If this equilibrium mixture is a darker colour at high temperatures and at low pressures, which of these statements about the reaction is correct?

- A. The reaction is exothermic and NO_2 is darker in colour than N_2O_4 .
 - B. The reaction is exothermic and N_2O_4 is darker in colour than NO_2 .
 - C. The reaction is endothermic and NO_2 is darker in colour than N_2O_4 .
 - D. The reaction is endothermic and N_2O_4 is darker in colour than NO_2 .
- 20** The K_{sp} of $\text{Ca}(\text{OH})_2$ at 25°C is 5.02×10^{-6} .

What is the molar solubility of calcium hydroxide at this temperature?

- A. 1.71×10^{-2}
- B. 1.08×10^{-2}
- C. 2.24×10^{-3}
- D. 5.06×10^{-4}

Chemistry

Section II – 80 marks

Attempt Questions 21-35

Allow about 2 hours and 25 minutes for this section.

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Show all relevant working in questions involving calculations.

Extra writing space is provided on pages 32 to 35. If you use this space, clearly indicate which question you are answering.

Question 21 (6 marks)

A class has been asked to conduct a chemical analysis of a common household substance for acidity or basicity.

Assess **TWO** different methods of how this analysis could be conducted.

(a)

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Question 22 (4 marks)

For the salt silver chloride, $K_{sp} = 1.77 \times 10^{-10}$ at 298 K.

A mixture is formed from 0.25 moles of AgNO_3 and 1.0 mole of NaCl in water.

(a) Explain the meaning of K_{sp} for silver chloride.

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(b) Calculate the concentration of silver ions, Ag^+ , at equilibrium in this mixture.

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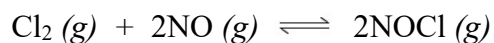
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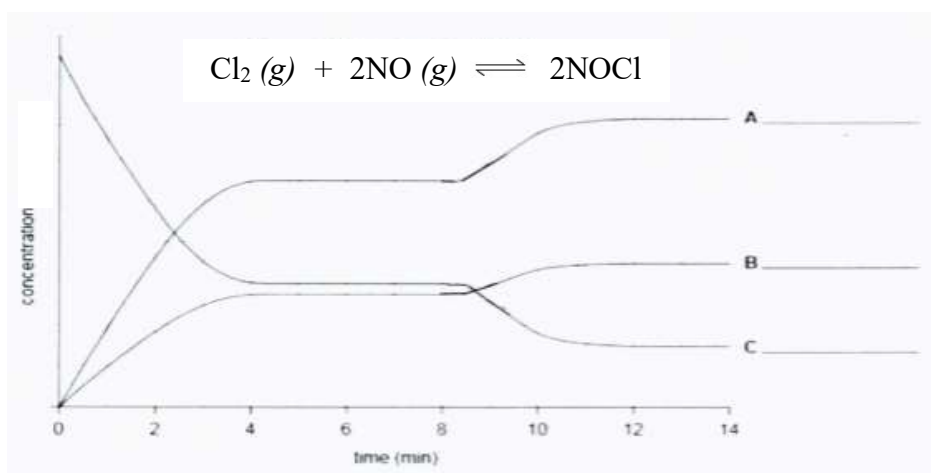
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Question 23 (5 marks)

Nitrogen monoxide, chlorine and NOCl form an equilibrium gaseous mixture:



The graph below shows the changes in concentration of the three species over time.



- (a) On the graph above, identify A, B and C as Cl₂, NO and NOCl. 1
- (b) Describe the change that was made to the reaction conditions at time t = 8.5 min. 1

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Question 23 continues on the next page

Question 23 (continued)

- (c) Will the equilibrium constant at $t = 12$ min be the same as at $t = 6$ min?
Explain your response.

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- (d) Will the presence of a catalyst impact on the equilibrium in any way?
Explain your response.

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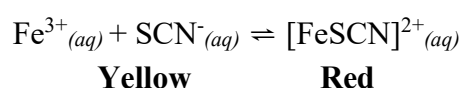
Question 24 (6 marks)

- (a) When an aqueous solution of iron (III) nitrate is added to an aqueous sodium phosphate solution, insoluble iron (III) phosphate is formed as one of the products. **1**

Write an equation that represents this reaction.

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- (b) If aqueous thiocyanate ions are added to an aqueous solution of iron (III) ions a red colour is produced as an iron (III) thiocyanate complex establishes an equilibrium, releasing heat.



- i. Explain why this equilibrium mixture appears red at room temperature. **1**

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- ii. Explain what would happen if a solution containing phosphate ions was added to the mixture. **2**

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- iii. Explain the effect of an increase in temperature on this equilibrium mixture. **2**

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Question 25 (5 marks)

Carbon dioxide gas reacts with carbon tetrafluoride gas to form an equilibrium with carbonyl fluoride gas (COF_2). The equilibrium constant for this reaction is 0.5.

- (a) Write the equation that represents this equilibrium reaction. 1

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- (b) Write the equilibrium constant expression for this reaction. 1

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- (c) A sample of 5.00 g of carbon dioxide gas and 29.00 g carbon tetrafluoride gas was injected into a sealed 5.00 L vessel. After a period of time the concentration of carbon tetrafluoride gas was measured again and found to be 4 g.L^{-1} . 3

Determine if this mixture is in equilibrium.

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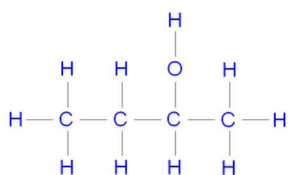
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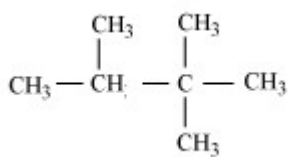
Question 26 (7 marks)

(a) Using IUPAC nomenclature, name the compounds shown below.

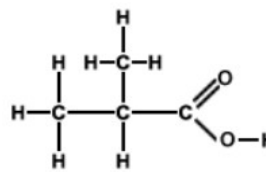
3



Compound 1



Compound 2



Compound 3

Compound 1

Compound 2

Compound 3

(b) Compare the intermolecular forces in the above 3 molecules and predict the order of boiling points (lowest to highest) of these molecules. Explain your prediction.

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Question 27 (5 marks)

A student was asked to distinguish between colourless organic liquids known to be

- butan-1-ol
- butan-2-ol
- 2-methylpropan-2-ol

- (a) Explain how the 3 isomers differ in their structures and how these differences allow them to be distinguished using a chemical test. Include a chemical equation in your answer.

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- (b) Outline the observations which would be made to allow the 3 liquids to be identified.

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Question 28 (2 marks)

Explain how Aboriginal and Torres Strait Islander Peoples have used the principles of solubility equilibria, when removing toxicity from foods such as cycad fruit.

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Question 29 (7 marks)

Various models of acids and bases have been used over time. Compare the Arrhenius theory, the Bronsted-Lowry theory and one other theory you have explored in your studies and explain the limitations of each model. **7**

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Question 30 (6 marks)

The table shows the acid dissociation constants at 25°C.

Acid	Formula	K_a
Acetic acid	CH_3COOH	1.8×10^{-5}
Chlorous acid	HClO_2	1.1×10^{-2}
Formic acid	HCOOH	1.8×10^{-4}
Hydrocyanic acid	HCN	6.2×10^{-10}
Hydrofluoric acid	HF	6.6×10^{-4}
Water	H_2O	1.0×10^{-14}
Lactic acid	$\text{CH}_3\text{CHOHCOOH}$	1.4×10^{-4}
Nitrous acid	HNO_2	7.2×10^{-4}
Phenol	$\text{C}_6\text{H}_5\text{OH}$	1.3×10^{-10}

- (a) Identify the strongest acid in the table and determine the $\text{p}K_a$ value for this acid.

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- (b) Calculate the pH of a 0.10 mol L^{-1} solution of nitrous acid.

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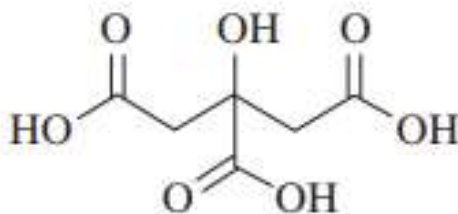
- (c) Calculate the percentage dissociation of a 0.1 mol L^{-1} solution of nitrous acid.

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Question 31 (4 marks)

A titration is carried out to determine the citric acid content of 1 L of orange juice. Citric acid, $C_6H_8O_7$, has the following structural formula and is a triprotic weak acid.



First, a solution of sodium hydroxide was standardised by titration against potassium hydrogen phthalate (a weak acid) and found to have a concentration of exactly $0.0540 \text{ mol L}^{-1}$.

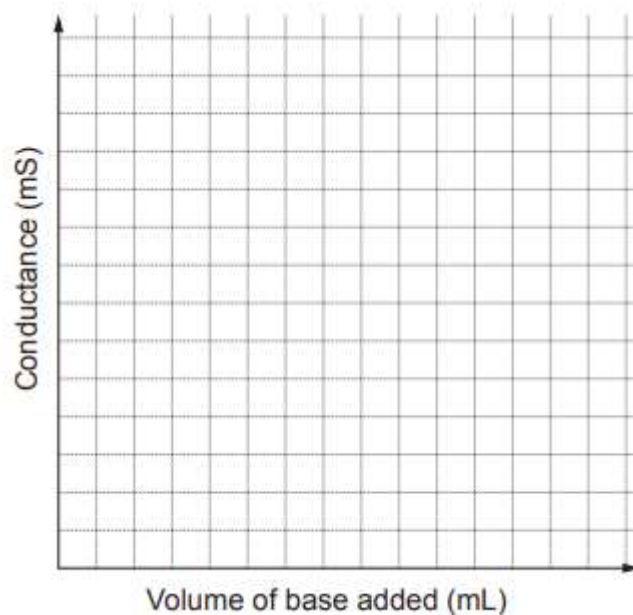
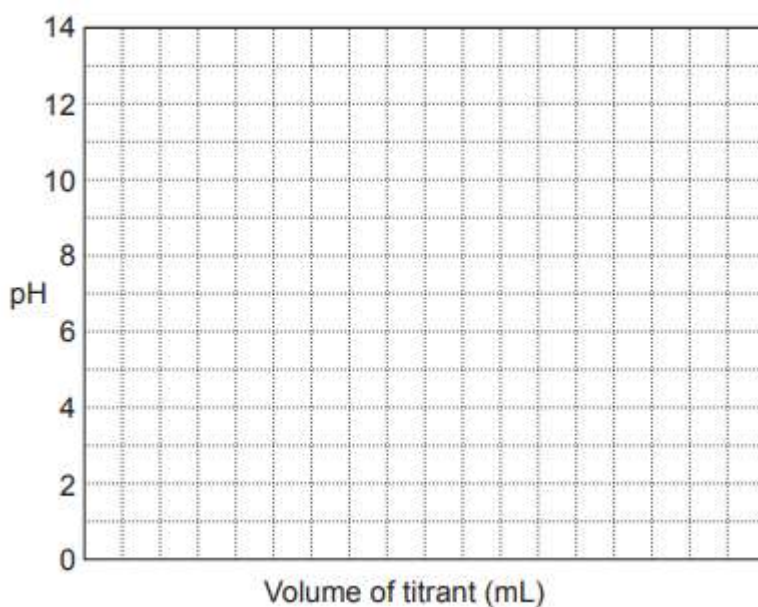
Next, the orange juice was diluted by taking 10 mL of juice and making it up to 50 mL with distilled water. 25 mL samples of the diluted orange juice were placed in clean conical flasks and titrated against the standardised sodium hydroxide. The average amount of sodium hydroxide needed to reach equivalence was 17.3 mL.

Question 31 continues on the next page

Question 31 (continued)

Complete the graphs to show the change in pH and the change in conductivity during the titration. Indicate the equivalence point on each graph.

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Question 32 (4 marks)

To determine the percentage by mass of calcium carbonate (CaCO_3) in an antacid tablet which had a mass of 0.250 g, the following procedure was used:

- The tablet was crushed and then placed in a beaker.
- A pipette was used to add 25.0 mL of 0.120 mol L^{-1} hydrochloric acid to the crushed tablet in the beaker.
- Once the reaction between the calcium carbonate and hydrochloric acid had stopped, phenolphthalein indicator was added to the reaction mixture.
- A burette was then used to add $0.0560 \text{ mol L}^{-1}$ sodium hydroxide to the beaker to neutralise the excess hydrochloric acid.
- The phenolphthalein changed from colourless to pink after 27.4 mL of the sodium hydroxide solution had been added.

- (a) Write a balanced equation for the reaction between calcium carbonate and hydrochloric acid. 1

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- (b) Determine the percentage by mass of calcium carbonate in the tablet. Show all working and reasoning. 3

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Question 33 (4 marks)

A green solution was made containing a weak acid, HA (which is a yellow molecule) and its conjugate base, A^- (which is blue).

- (a) Write an equation for the reaction which occurs when a strong base such as sodium hydroxide is added to this solution. **1**

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- (b) Write an equation for the reaction which occurs when a strong acid such as hydrochloric acid is added to this solution. **1**

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- (c) Use your equations from part (a) and (b) to explain why this solution can act as an indicator. **2**

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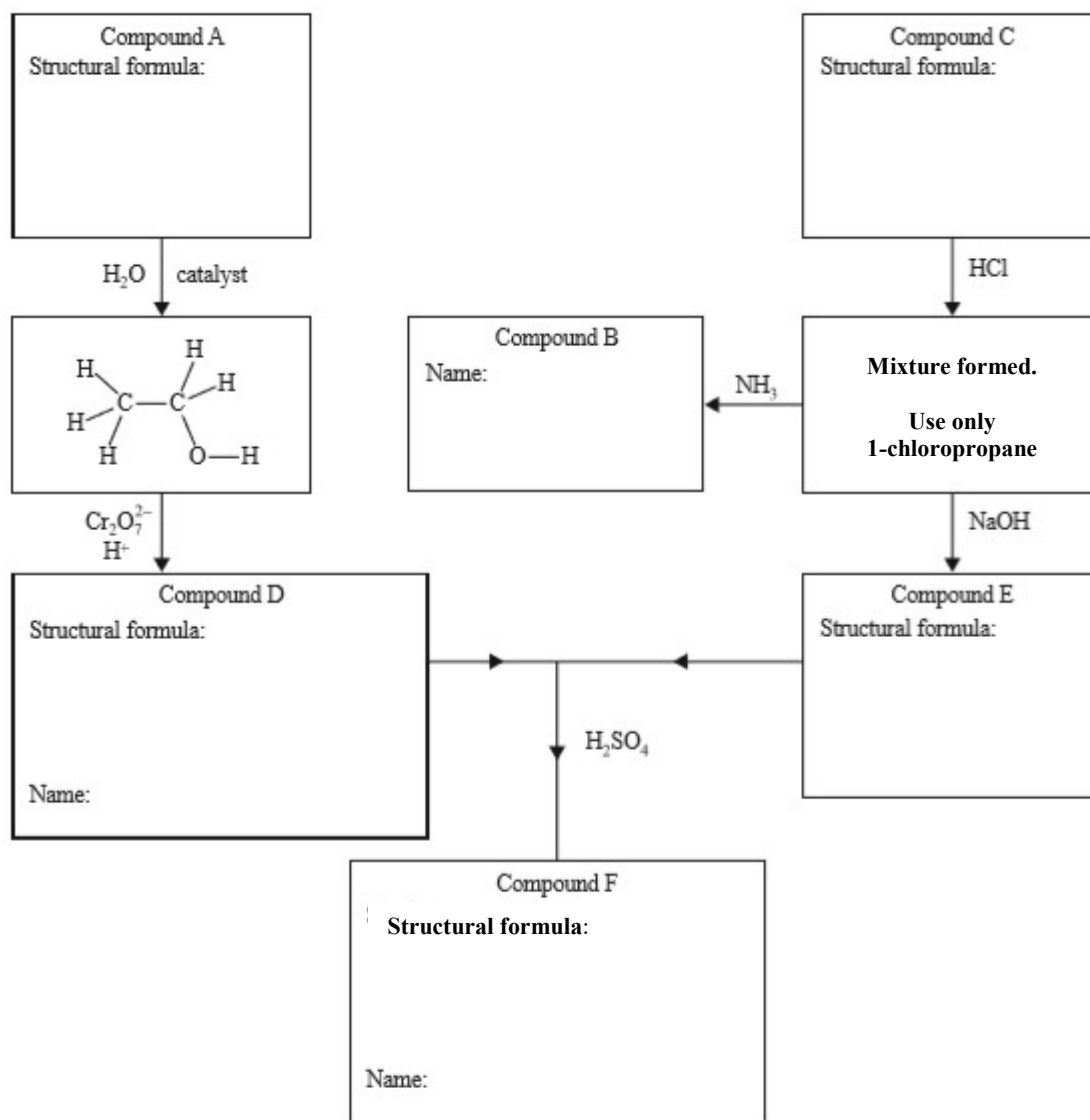
Question 34 (10 marks)

(a) Complete the diagram below by inserting:

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- structural formulae for compounds A, C, D, E and F
- the IUPAC names for compounds B, D and F

Compounds B and F may be synthesised as follows.



(b) Identify the types of reactions which occur when:

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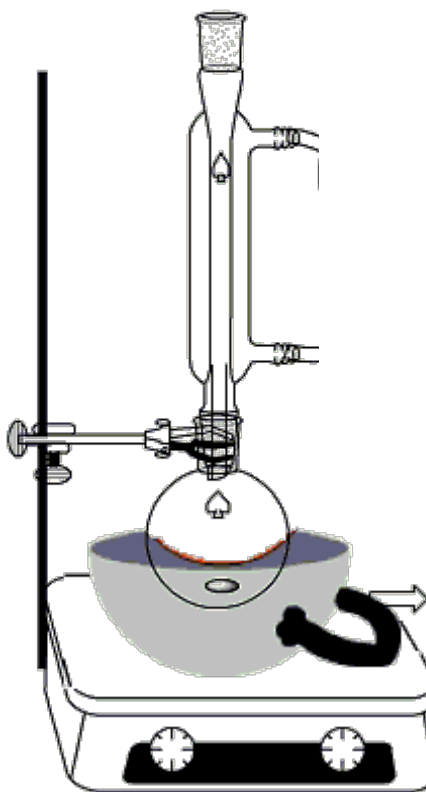
- Compound C is converted to 1-chloropropane
- 1-chloropropane is converted to Compound E

Question 35 (5 marks)

- (a) In the space below, draw the structural formula for each reactant required to produce pentyl acetate in the laboratory. Name each reactant. 2

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- (b) The image below shows the equipment used for this reaction. 3



Explain reasons for using the apparatus above in this technique.

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Space to answer Question 35 continues on the next page

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End of Paper

Section II - Extra writing space

If you use this space, clearly indicate which question you are answering.

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Student Name:.....

CHEMISTRY – MULTIPLE-CHOICE ANSWER SHEET

ATTEMPT ALL QUESTIONS

Question	1	A ☐	B ☐	C ☐	D ☐
	2	A ☐	B ☐	C ☐	D ☐
	3	A ☐	B ☐	C ☐	D ☐
	4	A ☐	B ☐	C ☐	D ☐
	5	A ☐	B ☐	C ☐	D ☐
	6	A ☐	B ☐	C ☐	D ☐
	7	A ☐	B ☐	C ☐	D ☐
	8	A ☐	B ☐	C ☐	D ☐
	9	A ☐	B ☐	C ☐	D ☐
	10	A ☐	B ☐	C ☐	D ☐
	11	A ☐	B ☐	C ☐	D ☐
	12	A ☐	B ☐	C ☐	D ☐
	13	A ☐	B ☐	C ☐	D ☐
	14	A ☐	B ☐	C ☐	D ☐
	15	A ☐	B ☐	C ☐	D ☐
	16	A ☐	B ☐	C ☐	D ☐
	17	A ☐	B ☐	C ☐	D ☐
	18	A ☐	B ☐	C ☐	D ☐
	19	A ☐	B ☐	C ☐	D ☐
	20	A ☐	B ☐	C ☐	D ☐

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Chemistry

FORMULAE SHEET

$$n = \frac{m}{MM}$$

$$q = mc\Delta T$$

$$pK_a = -\log_{10}[K_a]$$

$$c = \frac{n}{V}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$A = \epsilon lc = \log_{10} \frac{I_o}{I}$$

$$PV = nRT$$

$$\text{pH} = -\log_{10}[\text{H}^+]$$

Avogadro constant, N_A $6.022 \times 10^{23} \text{ mol}^{-1}$

Volume of 1 mole ideal gas: at 100 kPa and

at 0°C (273.15 K) 22.71 L

at 25°C (298.15 K) 24.79 L

Gas constant $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$

Ionisation constant for water at 25°C (298.15 K), K_w 1.0×10^{-14}

Specific heat capacity of water $4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

DATA SHEET

Solubility constants at 25°C

Compound	K_{sp}	Compound	K_{sp}
Barium carbonate	2.58×10^{-9}	Lead(II) bromide	6.60×10^{-6}
Barium hydroxide	2.55×10^{-4}	Lead(II) chloride	1.70×10^{-5}
Barium phosphate	1.3×10^{-29}	Lead(II) iodide	9.8×10^{-9}
Barium sulfate	1.08×10^{-10}	Lead(II) carbonate	7.40×10^{-14}
Calcium carbonate	3.36×10^{-9}	Lead(II) hydroxide	1.43×10^{-15}
Calcium hydroxide	5.02×10^{-6}	Lead(II) phosphate	8.0×10^{-43}
Calcium phosphate	2.07×10^{-29}	Lead(II) sulfate	2.53×10^{-8}
Calcium sulfate	4.93×10^{-5}	Magnesium carbonate	6.82×10^{-6}
Copper(II) carbonate	1.4×10^{-10}	Magnesium hydroxide	5.61×10^{-12}
Copper(II) hydroxide	2.2×10^{-20}	Magnesium phosphate	1.04×10^{-24}
Copper(II) phosphate	1.40×10^{-37}	Silver bromide	5.35×10^{-13}
Iron(II) carbonate	3.13×10^{-11}	Silver chloride	1.77×10^{-10}
Iron(II) hydroxide	4.87×10^{-17}	Silver carbonate	8.46×10^{-12}
Iron(III) hydroxide	2.79×10^{-39}	Silver hydroxide	2.0×10^{-8}
Iron(III) phosphate	9.91×10^{-16}	Silver iodide	8.52×10^{-17}
		Silver phosphate	8.89×10^{-17}
		Silver sulfate	1.20×10^{-5}

Aylward and Findlay, *SI Chemical Data* (5th Edition) is the principal source of data for this examination paper. Some data may have been modified for examination purposes.

Infrared absorption data

Bond	Wavenumber/ cm^{-1}
N—H (amines)	3300–3500
O—H (alcohols)	3230–3550 (broad)
C—H	2850–3300
O—H (acids)	2500–3000 (very broad)
$\text{C}\equiv\text{N}$	2220–2260
$\text{C}=\text{O}$	1680–1750
$\text{C}=\text{C}$	1620–1680
C—O	1000–1300
C—C	750–1100

^{13}C NMR chemical shift data

Type of carbon	δ/ppm
$\begin{array}{c} \quad \\ -\text{C}-\text{C}- \\ \quad \end{array}$	5–40
$\begin{array}{c} \\ \text{R}-\text{C}-\text{Cl or Br} \\ \end{array}$	10–70
$\begin{array}{c} \\ \text{R}-\text{C}-\text{C}- \\ \quad \\ \text{O} \end{array}$	20–50
$\begin{array}{c} \\ \text{R}-\text{C}-\text{N} \\ \quad \diagup \quad \diagdown \end{array}$	25–60
$\begin{array}{c} \\ -\text{C}-\text{O}- \\ \end{array}$ alcohols, ethers or esters	50–90
$\begin{array}{c} \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \end{array}$	90–150
$\text{R}-\text{C}\equiv\text{N}$	110–125
	110–160
$\begin{array}{c} \text{R}-\text{C}- \\ \\ \text{O} \end{array}$ esters or acids	160–185
$\begin{array}{c} \text{R}-\text{C}- \\ \\ \text{O} \end{array}$ aldehydes or ketones	190–220

UV absorption

(This is not a definitive list and is approximate.)

Chromophore	λ_{max} (nm)
C—H	122
C—C	135
C=C	162

Chromophore	λ_{max} (nm)
$\text{C}\equiv\text{C}$	173 178 196 222
C—Cl	173
C—Br	208

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{K}(s)$	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ba}(s)$	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ca}(s)$	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Na}(s)$	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Mg}(s)$	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	$\text{Al}(s)$	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Mn}(s)$	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(g) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Zn}(s)$	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Fe}(s)$	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ni}(s)$	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Sn}(s)$	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Pb}(s)$	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(g)$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(aq) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Cu}(s)$	0.34 V
$\frac{1}{2}\text{O}_2(g) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Cu}(s)$	0.52 V
$\frac{1}{2}\text{I}_2(s) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(aq) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Ag}(s)$	0.80 V
$\frac{1}{2}\text{Br}_2(l) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(aq) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(g) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(g) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(aq) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(g) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

PERIODIC TABLE OF THE ELEMENTS									
KEY									
Atomic Number		Symbol		Name		Standard Atomic Weight		79	
Symbol		Name		Standard Atomic Weight		79		Au	
Name		Standard Atomic Weight		79		Gold		197.0	
1		4		21		22		23	
1	H	2	He	3	Li	4	Be	5	B
1.008	Hydrogen	4.003	Helium	6.941	Lithium	9.012	Beryllium	10.81	Boron
3	Li	6.941	Lithium	9.012	Beryllium	10.81	Boron	12.01	Carbon
11	Na	12	Mg	13	Al	14	Si	15	P
22.99	Sodium	24.31	Magnesium	26.98	Aluminium	28.09	Silicon	30.97	Phosphorus
19	K	20	Ca	21	Sc	22	Ti	23	V
39.10	Potassium	40.08	Calcium	44.96	Scandium	47.87	Titanium	50.94	Vanadium
37	Rb	38	Sr	39	Y	40	Zr	41	Nb
85.47	Rubidium	87.61	Strontium	88.91	Yttrium	91.22	Zirconium	92.91	Niobium
55	Cs	56	Ba	57-71	Lanthanoids	72	Hf	73	Ta
132.9	Cesium	137.3	Barium	88-103	Actinoids	104	Rf	105	Db
87	Fr	88	Ra	89-103	Actinoids	104	Rf	105	Db
2	He	3	Li	4	Be	5	B	6	C
4.003	Helium	6.941	Lithium	9.012	Beryllium	10.81	Boron	12.01	Carbon
10	Ne	11	Na	12	Mg	13	Al	14	Si
20.18	Neon	22.99	Sodium	24.31	Magnesium	26.98	Aluminium	28.09	Silicon
18	Ar	19	K	20	Ca	21	Sc	22	Ti
39.95	Argon	39.10	Potassium	40.08	Calcium	44.96	Scandium	47.87	Titanium
36	Kr	37	Rb	38	Sr	39	Y	40	Zr
83.80	Krypton	85.47	Rubidium	87.61	Strontium	88.91	Yttrium	91.22	Zirconium
54	Xe	55	Cs	56	Ba	57-71	Lanthanoids	72	Hf
131.3	Xenon	132.9	Cesium	137.3	Barium	88-103	Actinoids	104	Rf
86	Rn	87	Fr	88	Ra	89-103	Actinoids	104	Rf
222	Radon	223	Francium	226	Radium	227-288	Actinoids	228	Ac
84	Po	85	At	86	Rn	87	Fr	88	Ra
209	Polonium	210	Astatine	211	Radon	212-288	Actinoids	213	Ac
127.6	Tellurium	127.6	Tellurium	127.6	Tellurium	127.6	Tellurium	127.6	Tellurium
52	Te	53	I	54	Xe	55	Cs	56	Ba
127.6	Tellurium	126.9	Iodine	131.3	Xenon	132.9	Cesium	137.3	Barium
51	Sb	52	Te	53	I	54	Xe	55	Cs
121.8	Antimony	127.6	Tellurium	126.9	Iodine	131.3	Xenon	132.9	Cesium
83	Bi	84	Po	85	At	86	Rn	87	Fr
209.0	Bismuth	209.0	Polonium	210	Astatine	211	Radon	212-288	Actinoids
115	Uup	116	Lv	117	Uus	118	Uuo	119	Uut
115	Uup	116	Lv	117	Uus	118	Uuo	119	Uut
114	Fl	115	Uup	116	Lv	117	Uus	118	Uuo
114	Fl	115	Uup	116	Lv	117	Uus	118	Uuo
113	Uut	114	Fl	115	Uup	116	Lv	117	Uus
113	Uut	114	Fl	115	Uup	116	Lv	117	Uus
112	Cn	113	Uut	114	Fl	115	Uup	116	Lv
112	Cn	113	Uut	114	Fl	115	Uup	116	Lv
111	Rg	112	Cn	113	Uut	114	Fl	115	Uup
111	Rg	112	Cn	113	Uut	114	Fl	115	Uup
110	Ds	111	Rg	112	Cn	113	Uut	114	Fl
110	Ds	111	Rg	112	Cn	113	Uut	114	Fl
109	Mt	110	Ds	111	Rg	112	Cn	113	Uut
109	Mt	110	Ds	111	Rg	112	Cn	113	Uut
108	Hs	109	Mt	110	Ds	111	Rg	112	Cn
108	Hs	109	Mt	110	Ds	111	Rg	112	Cn
107	Bh	108							

Elements with no reported values in the table have no stable nuclides.

Information on elements with atomic numbers 113 and above is sourced from the International Union of Pure and Applied Chemistry Periodic Table of the Elements (January 2016 version). The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of all other data. Some data may have been modified.



**2019 Higher School Certificate
Trial Examination**

Mapping Grid

Section I

Question	Answer
1	D
2	A
3	C
4	B
5	C
6	D
7	D
8	A
9	B
10	A
11	B
12	C
13	D
14	B
15	A
16	C
17	A
18	B
19	A
20	B

ANSWERS
MARKING
CRITERIA
ANNOTATED

Section II – 80 marks

Question 21 (7 marks)

21 (6 marks)

Outcomes Assessed: CH12–3, CH12–13

Targeted Performance Bands: 2–6

Criteria	Marks
• Describes TWO methods in detail-equipment and reagents AND • Outlines advantages AND disadvantages of each method AND • Provides overall judgement about the methods • Response is well written and succinct	6
• As above but only gives advantage and disadvantage of each method or response is not well written	5
• As above but methods are not described in enough detail- lacking chemicals and or equipment • Judgement is not explicit	4
• Outlines 2 methods, provides advantages or disadvantages of both methods or provides advantages and disadvantages of only 1 method	3
• Provides very basic details about one or 2 methods- no advantages or disadvantages	2
• Some correct details provided	1

Sample answer

The sample must be dissolved if a solid- it must be dissolved in distilled/deionised water. One method is to use indicators, so the sample should also form a transparent colourless solution when using indicators.

A couple of drops of the test substance should be placed on a well plate with 1-2 drops of universal indicator the colour change should be recorded and compared to the indicator's colour chart to identify the pH of the substance according to the colour chart.

This method is quick, easy to use and does not use much of the reagents, however it is destructive and it is reliant upon the person reading the colour accurately and comparing to chart accurately.

A pH probe can also be used which is non-destructive and very quick. The probe needs to be carefully calibrated and it needs to be used carefully, rinsed carefully and stored in appropriate buffer. It provides a digital reading of the pH often to 2 decimal places with a small error. Overall both methods allow the substance to be determined as acidic or basic depending on their pH the method would be chosen depending if the sample is abundant and if it needs decimal place results.

Students can make any judgement as long as their response supports their judgement

Question 22 (4 marks)

22 (a) (1 mark)

Outcomes Assessed: CH12-12**Targeted Performance Bands:** 5

Criteria	Marks
Correctly explains the meaning of K_{sp} by relating it to an equilibrium of ions in a saturated solution.	1

Sample answer

K_{sp} is the solubility product constant. It is the equilibrium constant for an equilibrium between a solid and its respective ions in a saturated solution. Its value indicates the degree to which a compound dissociates in water.

22 (b) (3 marks)

Outcomes Assessed: CH12-4, CH12-12**Targeted Performance Bands:** 5-6

Criteria	Marks
Correctly calculates the concentration of silver ions at equilibrium in the solution	3
Calculates a concentration of silver ions by referring to an excess of chloride ions and uses the correct K_{sp} expression	2
Calculates a concentration of silver ions by referring to an excess of chloride ions OR uses the correct K_{sp} expression	1

Sample answer

AgNO_3 is the limiting reagent, so moles AgCl formed = 0.25

$\therefore$ moles Cl^- remaining = 0.75

$$K_{sp} = [\text{Ag}^+] \times [\text{Cl}^-] = 1.77 \times 10^{-10}$$

$$[\text{Ag}^+] = \frac{1.77 \times 10^{-10}}{[\text{Cl}^-]} = \frac{1.77 \times 10^{-10}}{0.75} = 2.36 \times 10^{-10} \text{ mol L}^{-1}$$

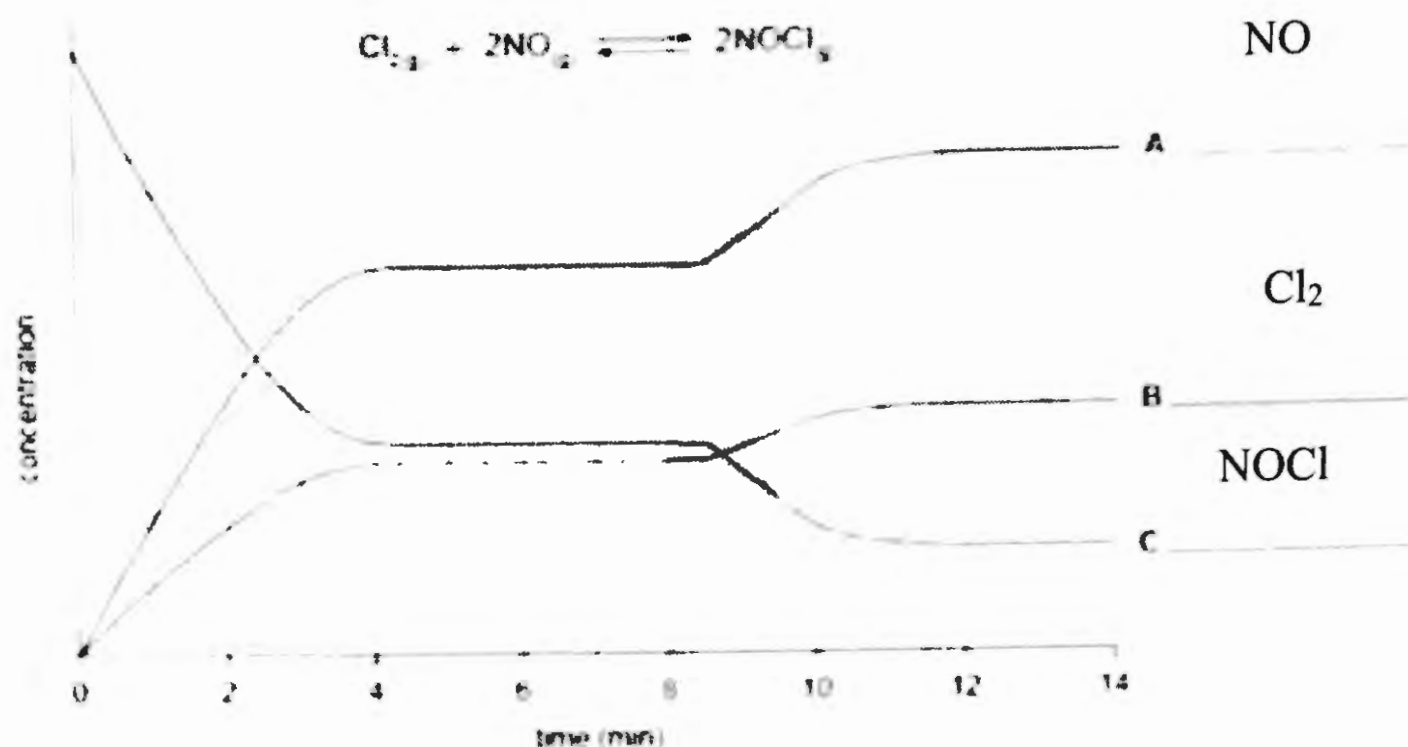
Question 23 (5 marks)

23 (a) (1 mark)

Outcomes Assessed: CH12-5, CH12-12**Targeted Performance Bands:** 3-4

Criteria	Mark
Correctly identifies A, B and C	1

Sample answer



23 (b) (1 mark)

Outcomes Assessed: CH12-7, CH12-12

Targeted Performance Bands: 3-5

Criteria	Mark
Describes that a temperature change has occurred (which could be an increase or decrease in temperature)	1

Sample answer

A change in temperature has occurred.

As no data has been given as to whether the forward reaction is exothermic or endothermic, it is not possible to state whether the temperature has increased or decreased. The change in temperature has caused a shift in the equilibrium towards the left (as the equation is written in the question paper) – producing higher concentrations of chlorine and nitrogen monoxide and a lower concentration of NOCl. If the forward reaction is exothermic, then an increase in temperature has occurred. Conversely, if the forward reaction (as written in the question paper) is endothermic, a decrease in temperature has occurred. Le Chatelier's Principle predicts that, if an equilibrium is disturbed, an increase in temperature favours the endothermic reaction.

23 (c) (2 marks)

Outcomes Assessed: CH12-6, CH12-7, CH12-12

Targeted Performance Bands: 2-3

Criteria	Marks
Identifies that the equilibrium constants at $t = 6$ min and at $t = 12$ min differ because the temperature has changed	2
Identifies that the equilibrium constants are different	1

Sample answer

Equilibrium constants depend on temperature.

Since the temperature changed at $t = 8$ min but did not change again, the equilibrium constant at $t = 6$ min will differ from the equilibrium constant $t = 12$ min.

23 (d) (1 mark)

Outcomes Assessed: CH12-7, CH12-12

Targeted Performance Bands: 3-4

Criteria	Mark
• Correct answer in terms of rates and yield	1

Sample answer

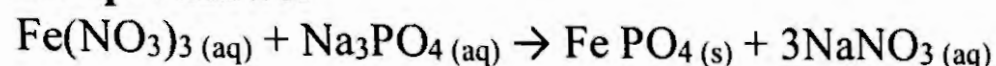
A catalyst will increase the rate both of the backward and forward reactions equally and will hence decrease the time taken for a new equilibrium to be achieved after an imposed change. However, it will not alter the concentrations of the species at equilibrium. Hence the presence of a catalyst will not alter the equilibrium constants.

Question 24 (5 marks)

24 (a) (1 mark)

Outcomes Assessed: CH12-4, CH12-15**Targeted Performance Bands:** 4

Criteria	Mark
• Equation has correct formulae, is correctly balanced and has correct states.	1

Sample answer

24 (b)(i) (2 marks)

Outcomes Assessed: CH12-4, CH12-6, CH12-12**Targeted Performance Bands:** 4-5

Criteria	Marks
• Clear and accurate explanation of why the equilibrium mixture is reddish at room temperature with reference to Le Chatelier's Principle	2
• Explanation of why the equilibrium mixture is reddish at room temperature with some detail missing.	1

Sample answer

At equilibrium this reaction is reddish because even though all three ions are present in the solution, the iron (III) thiocyanate ions have the highest concentration. The equilibrium lies to the right at this temperature.

24 (b)(ii) (2 marks)

Outcomes Assessed: CH12-4, CH12-6, CH12-12**Targeted Performance Bands:** 4-6

Criteria	Marks
• Clearly and accurately explains that the addition of phosphate ions would precipitate a solid and cause a shift in the equilibrium.	2
• Explains that the addition of phosphate ions would precipitate a solid and cause a shift in the equilibrium with some detail missing	1

Sample answer

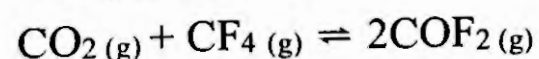
If phosphate ions were added to this mixture they would react with the Fe^{3+} ions and form a solid. Since these ions are now removed from the solution the equilibrium would shift to the left and the mixture would become a lighter red or yellowish colour.

Question 25 (5 marks)

25 (a) (1 mark)

Outcomes Assessed: CH12-4, CH12-12**Targeted Performance Bands:** 2-3

Criteria	Mark
• Equation is an equilibrium, has correct formulae and is correctly balanced.	1

Sample answer

25 (b) (1 mark)

Outcomes Assessed: CH12-5, CH12-7, CH12-12

Targeted Performance Bands: 2-3

Criteria	Mark
• Correctly writes the equilibrium constant expression for this reaction.	1

Sample answer

$$K = \frac{[\text{COF}_2]^2}{[\text{CO}_2] \times [\text{CF}_4]}$$

25 (c) (3 marks)

Outcomes Assessed: CH12-4, CH12-5, CH12-6, CH12-12

Targeted Performance Bands: 5-6

Criteria	Marks
• Correctly determines that the mixture is not in equilibrium by correctly calculating concentrations and using an ICE table or equivalent.	3
• Determines that the mixture is not in equilibrium by correctly calculating concentrations OR using an ICE table or equivalent.	2
• Correctly calculating concentrations or uses an ICE table or a correct mathematical expression for K.	1

Sample answer

$$\text{Initial } [\text{CO}_2] = 1.0 \text{ g L}^{-1} = 1 / (12.01 + 32) \text{ mol L}^{-1} = 0.0227221086 \text{ mol L}^{-1}$$

$$\text{Initial } [\text{CF}_4] = 5.8 \text{ g L}^{-1} = 5.8 / [12.01 + (4 \times 19)] \text{ mol L}^{-1} = 0.06590160209 \text{ mol L}^{-1}$$

$$\text{Final } [\text{CF}_4] = 4 \text{ g L}^{-1} = 4 / [12.01 + (4 \times 19)] \text{ mol L}^{-1} = 0.04544938075 \text{ mol L}^{-1}$$

	[CO ₂]	[CF ₄]	[COF ₂]
Initial	0.023	0.066	0.00
Change	-0.021	-0.021	+0.042
Final	0.002	0.045	0.042

$$K_{\text{eq}} = \frac{[\text{COF}_2]^2}{[\text{CO}_2] \times [\text{CF}_4]} = \frac{(0.042)^2}{0.002 \times 0.045} = 19.6$$

K_{eq} is greater than 0.5. The mixture is not in equilibrium.

Question 26 (7 marks)

26 (a) (3 marks)

Outcomes Assessed: CH12-7, CH12-14

Targeted Performance Bands: 3-5

Criteria	Marks
• Names THREE compounds correctly	3
• Names TWO compounds correctly	2
• Names ONE compound correctly	1

Sample answer

Compound 1 = butan-2-ol
 Compound 2 = 2,2,3-trimethylbutane
 Compound 3 = 2-methylpropanoic acid

23 (b) (4 marks)

Outcomes Assessed: CH12-7, CH12-14

Targeted Performance Bands: 3-5

Criteria	Marks
- Predicts the correct order of boiling points - Explains thoroughly the impact of the different intermolecular forces - Identifies that Compound 1 has hydrogen bonding as the strongest intermolecular force - Identifies that Compound 2 has dispersion (temporary dipole-dipole forces) only - Identifies that Compound 3 has hydrogen bonding (2 hydrogen bonds form between adjacent molecules)	4
- Predicts the correct order of boiling points - Explains thoroughly the impact of the different intermolecular forces - Identifies the intermolecular forces in 2 of the 3 compounds	3
- TWO of: - Predicts the correct order of boiling points - Explains thoroughly the impact of the different intermolecular forces - Identifies the intermolecular forces in 2 of the 3 compounds	2
- ONE of: - Predicts the correct order of boiling points - Explains thoroughly the impact of the different intermolecular forces - Identifies the intermolecular forces in 1 of the 3 compounds	1

Sample answer

The order of increasing boiling point is Compound 2, Compound 1, Compound 3.

The stronger the intermolecular forces, the higher the boiling point, as greater energy is needed to separate the liquid molecules to form a gas.

Compound 2 is non-polar and has only weak intermolecular forces (dispersion or temporary dipole-dipole forces) caused by the electrical interaction of molecules as they collide (protons from 1 molecule being attracted to electrons from the other as the molecules are temporarily distorted on collision).

Compound 1 is polar and would experience hydrogen bonding forces, as well as weaker temporary and permanent dipolar forces) as molecules interact. These are strong intermolecular forces where the electronegativity of the oxygen results in a very polar bond with hydrogen in the O–H group. This hydrogen is attracted to the oxygen of a neighbouring alcohol molecule. The geometry of the molecule only allows 1 H-bond per pair of molecules at any instant.

Compound 3 is an acid and has the very polar –COOH functional group. The hydrogen atom of the –COOH can form a hydrogen bond with an oxygen of the neighbouring acid molecule. The planar nature of this –COOH group allows 2 H-bonds per pair of molecules. Hence the intermolecular forces and thus boiling points are highest in Compound 3 and lowest in Compound 2.

Question 27 (5 marks)

27 (a) (3 marks)

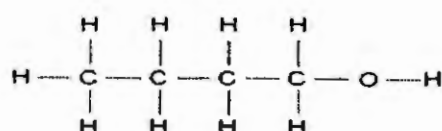
Outcomes Assessed: CH12-7, CH12-15**Targeted Performance Bands:** 2-5

Criteria	Marks
- Identifies the isomers as primary, secondary or tertiary alcohols and hence explains the difference in structures of the 3 isomers - Outlines the method of distinguishing between the 3 alcohols, which depends on the difference in structures	2
ONE of the above	1

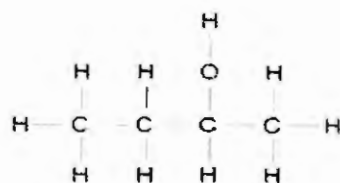
Sample answer

The 3 chemicals listed are isomers (all have the same molecular formula) and are all alcohols. They differ, in that butan-1-ol is a primary alcohol, butan-2-ol is a secondary alcohol and 2-methylpropan-2-ol is a tertiary alcohol.

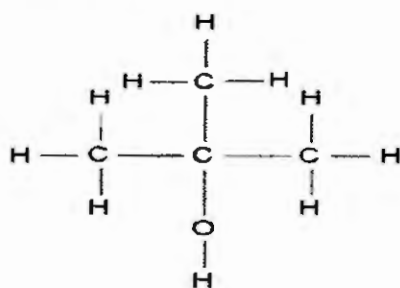
Primary alcohol: the –OH group is attached to a terminal carbon. The –OH group is attached to a carbon which is attached to only 1 other carbon; e.g. butan-1-ol.



Secondary alcohol: the –OH group is attached to a carbon which is attached to 2 other carbons; e.g. butan-2-ol.



Tertiary alcohol: the –OH group is attached to a carbon which is attached to 3 other carbons; e.g. 2-methylpropan-2-ol.



The 3 types of alcohols are distinguished between by reaction of a sample of each alcohol with acidified potassium dichromate solution, followed by testing of any products to distinguish between the primary and secondary alcohols. The 3 alcohols react differently; primary alcohols produce carboxylic acids, secondary alcohols produce alkanones and tertiary alcohols do not react.

27 (b) (2 marks)

Outcomes Assessed: CH12-7, CH12-15

Targeted Performance Bands: 2-4

Criteria	Marks
• Outlines the observations (being colour changes from orange to green) for the oxidation reactions of primary and secondary alcohols AND outlines that no colour change is observed for the tertiary alcohol AND outlines that the product of oxidation of a primary alcohol is acidic and hence can be distinguished from the product of oxidation of a secondary alcohol	2
• ONE of the above	1

Sample answer

Primary alcohols react with excess acidified potassium dichromate solution to form a carboxylic acid. The oxidising agent is converted into green chromate ions. The presence of the acid in solution could be tested by **separating the product** and testing a sample of the product with sodium carbonate solution; carbon dioxide bubbles would be produced. In this reaction, butan-1-ol is converted to butanal and then butanoic acid.

Secondary alcohols react with acidified potassium dichromate solution but form ketones/alkanones, not acids. Here butan-2-ol reacts to form propanone (acetone). There is a colour change but no acid is formed.

Tertiary alcohols are not oxidised by acidified potassium dichromate solution, so no colour change occurs when the reagents are mixed.

Question 28 (2 marks)

28 (2 marks)

Outcomes Assessed: CH12-7, CH12-12

Targeted Performance Bands: 2-4

Criteria	Marks
• Explains thoroughly how Aboriginal and Torres Strait Islander Peoples use the principles of solubility equilibria to remove toxins from foods	2
• Explains some aspects of removal of toxicity by leaching	1

Sample answer

Aboriginal and Torres Strait Islander Peoples remove soluble toxic substances from foods by leaching.

For example, cycad seeds are used in making breads, but these breads are poisonous if the seeds are not crushed and leaching used to remove the poisons.

The paste formed is left in water until the solubility of the poison in water is reached. This is achieved when the solubility equilibrium between the solid poison and the solution of poison has been achieved. The solution has then become saturated with the toxin. The equilibrium is then destroyed by removing the saturated solution containing the toxin, adding new pure water and waiting until equilibrium is again reached. The saturated solution is removed, fresh water added and the process repeated. Each step drives the equilibrium to the right and reduces the amount of toxin remaining in the seed.

The seed is then roasted to remove any residual toxin, before being used in cooking.

Question 29 (7 marks)

Criteria	Marks
- Explains Arrhenius AND Bronsted Lowry AND one other theory with correct examples provided - Distinguishes between the theories/makes clear how they advanced our understanding - Answer is concise and logical	7
As above but answer is not set out in a logical manner (jumps from one model to another in no order) <i>or as 7 but missing some detail.</i>	6
- Explains Arrhenius AND Bronsted Lowry AND one other theory with correct examples provided of at least 2 theories - Response acknowledges our understanding has changed with each model but does not make clear how this has occurred.	5
Explains Arrhenius AND Bronsted Lowry AND one other theory with <u>correct</u> examples provided of at least 2 theories <i>or as above but missing exs.</i>	4
Identifies and defines in very brief detail 3 theories succession of understanding not demonstrated <i>or no/limited examples or incorrect grasp.</i>	3
Identifies and defines in very brief detail 2 theories succession of understanding not demonstrated	2
Some correct information provided	1

Sample answer

The Brönsted-Lowry theory classifies an acid as a proton donor and a base as a proton acceptor. For example the proton is lost from the acid HCl (g) and gained by the base ammonia. The Brönsted-Lowry theory classification is irrespective of the state of the reactants, so the reaction can occur in the gaseous state.

Provide a correct equation

Arrhenius described acid-base reactions as reactions which took place only in aqueous solution. An acid produced H^+ (now H_3O^+) as the only positive ion in solution. A base was an oxide or hydroxide that produced hydroxide ions in aqueous solution. Ammonia would not be classified as a base with this theory for example.

Provide correct examples eg HCl, NaOH

Another theory described correctly with examples/equations

Our understanding has changed with each model as new information is gained our classification of what an acid and base is has changed/improved.

Question 30 (6 marks)

(a) (2 marks)

Outcomes Assessed: CH12-5, CH12-13

Targeted Performance Bands: 2-4

Criteria	Marks
Identifies chlorous acid as the strongest acid AND determines the pK_a value for the identified acid	2
Identifies chlorous acid as the strongest acid OR determines the pK_a value for the identified acid	1

Not doing sig fig for pH

Sample answer

Chlorous acid, HClO_2

$\text{pK}_a = -\log(\text{K}_a) = -\log(1.1 \times 10^{-2}) = 1.96$ (2 s.f.) (same rule for s.f. in pK_a calculations as in pH)

27 (b) (3 marks)

Outcomes Assessed: CH12-5, CH-6, CH12-13

Targeted Performance Bands: 3-6

Criteria	Marks
- Calculates the pH to 2 s.f. - Correctly determines the $[\text{H}_3\text{O}^+]$ - Shows correct working and reasoning (-x can ignore so small)	3
- Correctly determines the $[\text{H}_3\text{O}^+]$ - Shows correct working and reasoning	2
Shows some correct working and reasoning	1

Sample answer



$$\text{K}_a \text{HNO}_2 = \frac{[\text{NO}_2^-][\text{H}_3\text{O}^+]}{[\text{HNO}_2]} = 7.2 \times 10^{-4}$$

Let x moles of HNO_2 ionise, forming x moles of H_3O^+

$$\text{K}_a \text{HNO}_2 = \frac{[x][x]}{[0.10 - x]} = 7.2 \times 10^{-4}$$

Since x will be small by comparison with 0.10

$$\text{Hence } [x]^2 = 7.2 \times 10^{-4} \times 0.10 = 7.2 \times 10^{-5}$$

$$[\text{H}_3\text{O}^+] = \sqrt{7.2 \times 10^{-5}} = 8.5 \times 10^{-3} \text{ mol/L}$$

$$\text{Hence pH} = -\log_{10}(8.5 \times 10^{-3}) = 2.07$$

Note for teachers: In mathematical terms, the number to the left of the decimal point in a logarithm is called the characteristic and the number to the right of the decimal point is called the mantissa. The mantissa has as many significant figures as the number from which the logarithm was determined. Hence $[\text{H}_3\text{O}^+] = 8.5 \times 10^{-3}$, $\text{pH} = 2.07$.

30 (c) (1 mark)

Outcomes Assessed: CH12-5, CH-6, CH12-13

Targeted Performance Bands: 2-3

Criteria	Mark
Correctly calculates the percentage dissociation of nitrous acid	1

Sample answer

$$[\text{H}_3\text{O}^+] = 8.5 \times 10^{-3} \text{ mol/L}$$

$$[\text{HNO}_2] = 0.10 \text{ mol/L}$$

$$\% \text{ dissociation} = \frac{8.5 \times 10^{-3}}{0.10} \times 100$$

$$\% \text{ dissociation} = 8.5\% \text{ (2sf)}$$

should be 1 sig fig.

Sample
Q's

Question 31 (4 marks)

Outcomes Assessed: CH12-4, CH-6, CH12-13

Targeted Performance Bands: 2-5

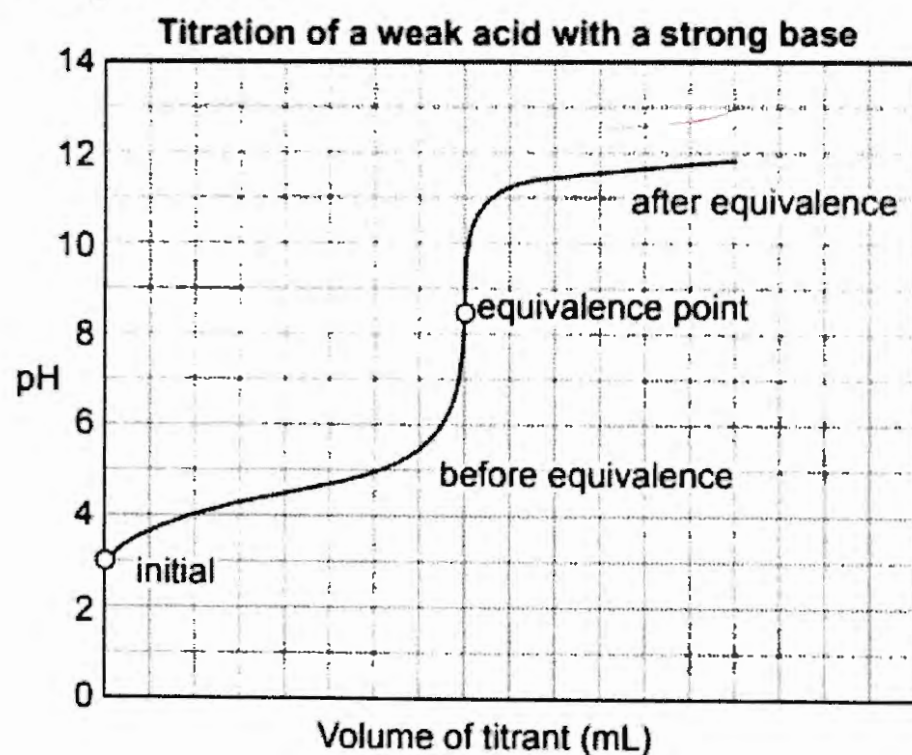
weak acid + strong base
pH begins/ends approp.

Marking guidelines (b):

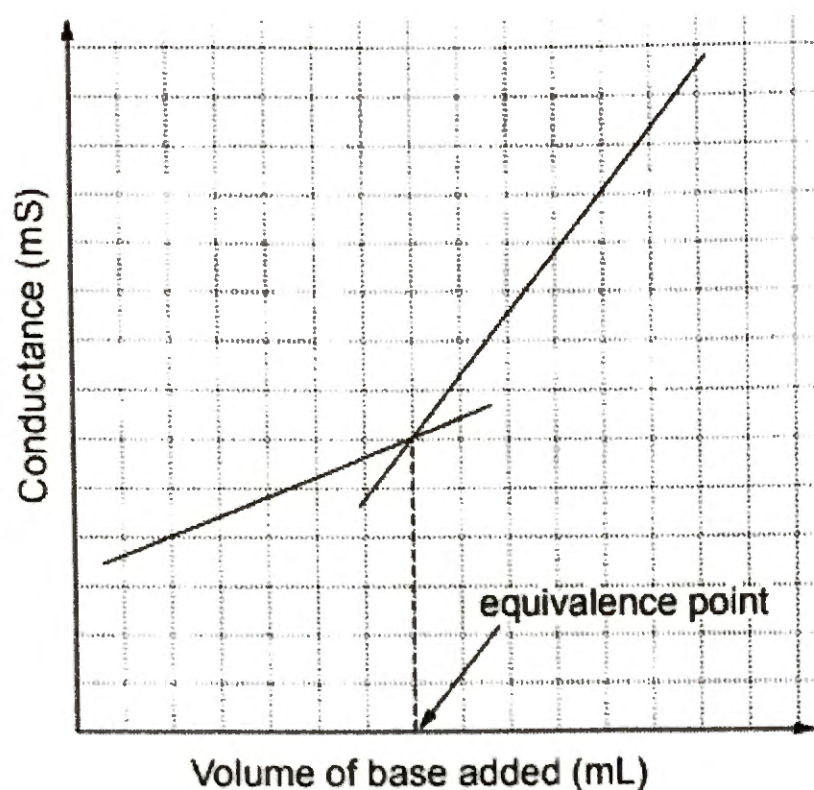
Criteria	Marks
Provides TWO correct graphs with equivalence point indicated for each	4
- Provides TWO substantially correct graphs with an equivalence point indicated OR - Provides TWO correct graphs	3
- Provides a substantially correct graph with an equivalence point indicated OR - Provides TWO substantially correct graphs	2
Provides some relevant information	1

Eg. basically
correct titration
curve in
reverse.
But not for
conductance
as labelled
vol. of base

Sample answer



• initial + final
pH must be
appropriate
• equiv. should be
suitable position
• shape reasonable
of curve.
• equiv. labelled/
indicated.



beginning low as weak acid
 • after equiv. higher & go up signif.

this OK too, as some textbooks use this, and can be complicated depending on subst + conc.

Question 32 (7 marks)

28 (a) (1 mark)

Outcomes Assessed: CH12-7, CH12-13**Targeted Performance Bands:** 2-3

Criteria	Mark
• Writes a correct balanced equation	1

Sample answer

28 (b) (3 marks)

Outcomes Assessed: CH12-5, CH-6, CH12-13**Targeted Performance Bands:** 2-5

Criteria	Marks
• Calculates correctly the percentage by mass of CaCO_3 in the tablet	3
• Determines the moles of HCl in excess	2
• Determines the moles of HCl added to the flask	1

Sample answer

There were 3 steps in the process:

- A tablet was crushed and placed in a beaker
- 25.0 mL of 0.120 M hydrochloric acid was pipetted into the beaker
- The excess HCl was titrated with NaOH to determine the no. of moles of HCl in excess and hence the no. of moles of CaCO_3

Moles HCl added initially to react and dissolve $\text{CaCO}_3 = cV = 0.120 \times 0.025 = 0.00300 \text{ mol}$ Moles NaOH required for titration excess HCl = $0.0274 \times 0.0560 = 0.00153 \text{ mol}$

or 0.0015344

Hence moles of excess HCl = 0.00153 (as NaOH and HCl react in 1:1 ratio)

Moles of HCl which had reacted with tablet = $0.00300 - 0.00153 = 0.00147 \text{ mol}$

0.0014656

Moles $\text{CaCO}_3 = \frac{1}{2}$ moles HCl (from balanced equation)Hence moles CaCO_3 in tablet = 0.000735 mol7.325 x 10⁻⁴Mass of CaCO_3 in tablet = $n \times M = 0.000735 \times 100.1 = 0.0736 \text{ g (3 s.f.)}$ % CaCO_3 in tablet = $0.0736/0.250 \times 100 = 29.4\%$

29.4

(was fairly kind on labels, but do not ignore in HSC!)(keep sig fig eg 0.00300 mol not 0.003 mol)
initial HCl

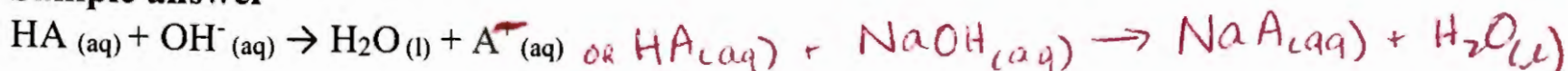
(keep working clear, show detail of your process in case you make an error somewhere)

Question 33 (4 marks)

33 (a) (1 mark)

Outcomes Assessed: CH12-5, CH12-13**Targeted Performance Bands:** 3

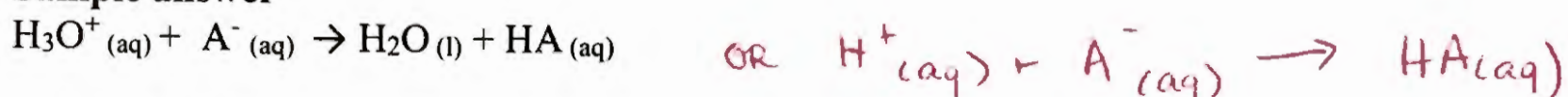
Criteria	Mark
• Correct equation	1

Sample answer

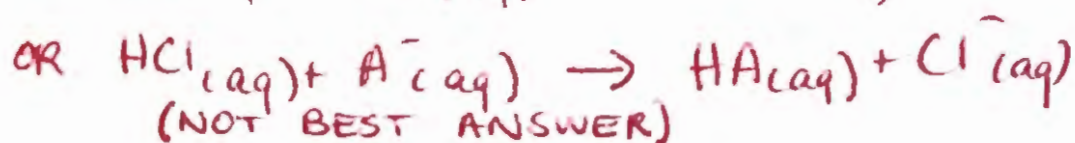
33 (b) (1 mark)

Outcomes Assessed: CH12-5, CH12-13**Targeted Performance Bands:** 3

Criteria	Mark
• Correct equation	1

Sample answer

33 (c) (3 marks)

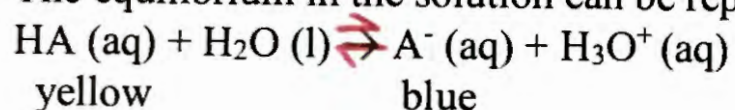
Outcomes Assessed: CH12-5, CH12-13**Targeted Performance Bands:** 3-5

Marking guidelines	Marks
- Explains why the solution can act as an indicator, as the colour changes with changes in pH / changes in concentration of H_3O^{+} - AND - Refers to the reactions in parts (a) and (b)	2
Explains why the solution can act as an indicator, as the colour changes with changes in pH / changes in concentration of H_3O^{+}	1

Sample answer

The solution is a green equilibrium mixture of HA (yellow) and A⁻ (blue).

The equilibrium in the solution can be represented:



When a strong base is added, the reaction in (a) occurs and the indicator solution turns blue.

Considering the equilibrium reaction above, the green solution turns blue, as the concentration of H_3O^{+} decreases and the indicator equilibrium shifts to the right.

When a strong acid is added, the reaction in (b) occurs. Considering the equilibrium reaction above, the green solution turns yellow, as the equilibrium shifts to the left as the concentration of H_3O^{+} increases.

Hence the green solution can act as an indicator as it is a different colour in acidic and basic solutions; i.e. in solutions of different pH.

Question 34 (10 marks)

34 (a) (8 marks)

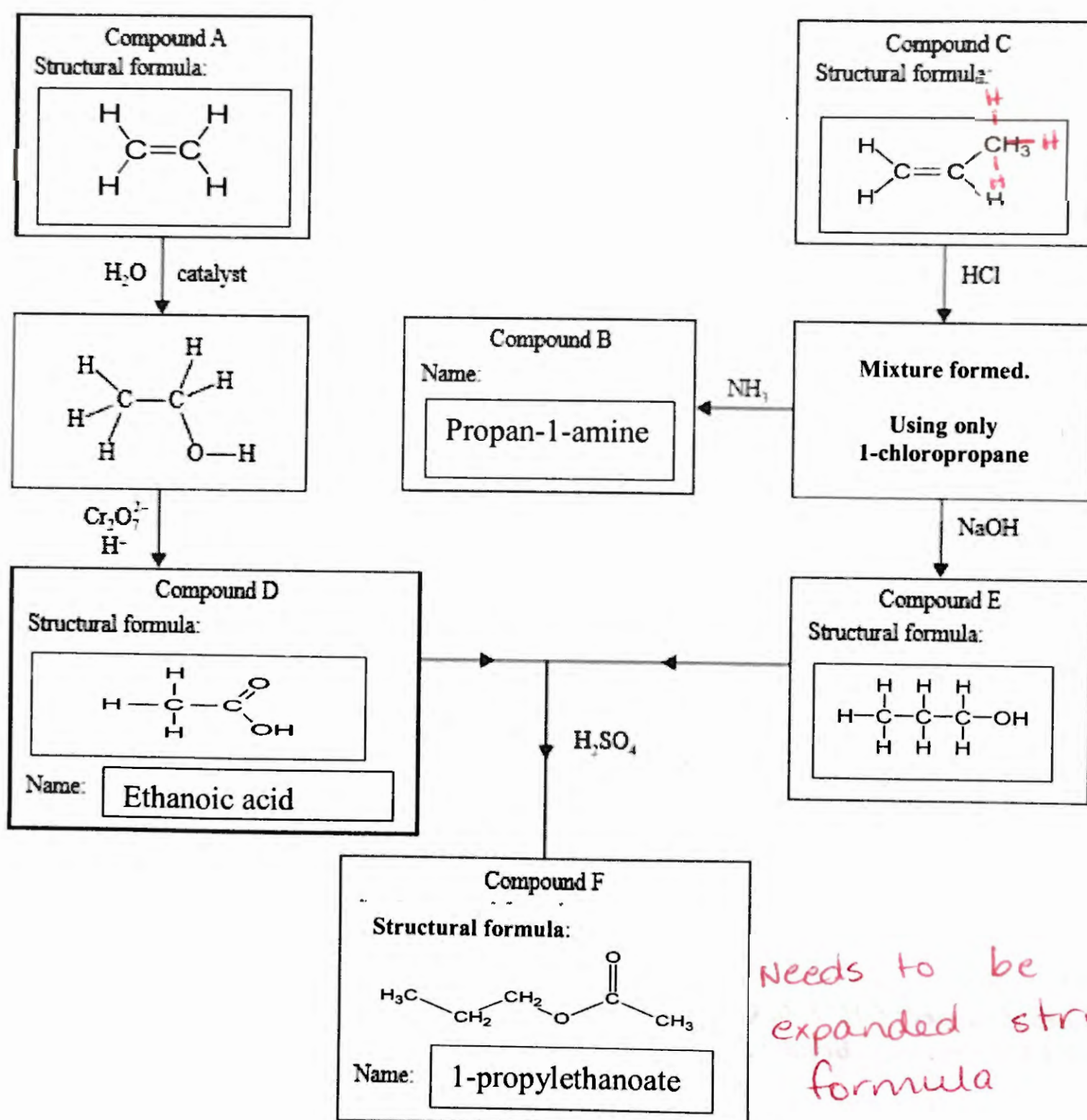
Outcomes Assessed: CH12-5, CH12-7, CH12-14**Targeted Performance Bands:** 3

Criteria	Marks
• Completes 8 structural formulae or names	8

• Completes 7 structural formulae or names	7
• Completes 6 structural formulae or names	6
• Completes 5 structural formulae or names	5
• Completes 4 structural formulae or names	4
• Completes 3 structural formulae or names	3
• Completes 2 structural formulae or names	2
• Completes 1 structural formula or name	1

Sample answer

Compounds B and F may be synthesised as follows.



34 (b) (2 marks)

Outcomes Assessed: CH12-7, CH12-14

Targeted Performance Bands: 4-5

Criteria	Marks
• TWO correct answers	2
• ONE correct answer	1

Sample answer

- Compound C is converted to 1-chloropropane in an **addition** reaction (Note for teachers: the addition reaction produces a mixture of 1-chloropropane and 2-chloropropane. 1-chloropropane is the minor product (see Markovnikoff's Rule).)
- 1-chloropropane is converted to Compound E in a **substitution** reaction.

Question 35 (5 marks)

32(a) (2 marks)

Outcomes Assessed: CH12-6, CH12-7, CH12-14

Targeted Performance Bands: 4-6

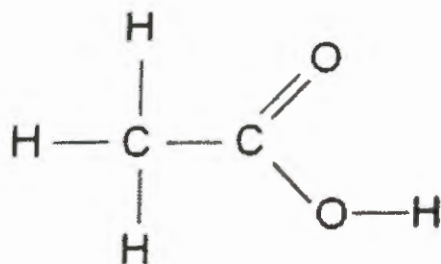
Marking Guidelines	Marks
• Shows structural formulas for 1-pentanol AND acetic acid <i>with correct name</i>	2
• Shows a structural formula for 1-pentanol OR acetic acid	1

Sample answer

pentan-1-ol



ethanoic acid (acetic acid)



35(b) (3 marks)

Outcomes Assessed: CH12-6, CH12-12, CH12-14

Targeted Performance Bands: 3-4

Marking Guidelines	Marks
• Thoroughly explains TWO reasons for using the reflux technique	3

• Explains TWO reasons for using the reflux technique with some detail missing	2
• Explains ONE reason for using the reflux technique	1

Sample answer

The reaction mixture is boiled under reflux, using an open condenser. This enables the reaction to be carried out at the boiling point, to maximise rate, without loss of volatile reactants or products via vaporisation. The use of an open condenser avoids any hazardous build-up of pressure in the system.

- Important to identify the equipment.

Don't use : cooling tube) etc.
heating mat

- Alkanols are the most volatile species in the reaction